

Tarikul Islam Tamiti
[Email](#) [Scholar](#) [LinkedIn](#)

EDUCATION

INFT PhD program with CYSE Concentration August 2024 (Ongoing) August 2024-Present
Department of Cyber Security Engineering, George Mason University, Fairfax, Virginia
- GPA: 3.88/4.00
Bachelor of Science in Electrical & Computer Engineering (ECE) October 2022
Rajshahi University of Engineering & Technology (RUET), Bangladesh
- GPA: 3.98/4.00, Highest GPA in the 2016-17 series (equivalent to valedictorian of class of 2021)

PAPERS

-Tamiti, T. I., & Barua, A. (2025). *Nonlinear framework for speech bandwidth extension* ([arXiv:2510.01109](#)). Accepted to **ICASSP 2026**.
-Tamiti, T. I., Joshi, B., Hasan, R., & Barua, A. (2025, October 1). *HVAC-EAR: Eavesdropping human speech using HVAC systems* ([arXiv:2510.01082](#)). Accepted to **ICASSP 2026**.
-Tamiti, T. I., Joshi, B., Hasan, R., & Barua, A. (2025). Reconstructing intelligible speech from the pressure sensor data in HVACs ([arXiv:2506.22311](#)). Accepted to **HOST 2026**.
-Tamiti, T. I., Mamun, N., & Barua, A. (2025). *CIS-BWE: Chaos-informed speech bandwidth extension* ([arXiv:2510.01109](#)). **ACL ARR October 2025 submission (Meta-review: 4/5)**.
-Sengupta, S., Ghosh, S., Mitra, P., & Tamiti, T. I. (2024). *Milestones in Bengali sentiment analysis leveraging transformer-models: Fundamentals, challenges and future directions* ([arXiv:2401.07847](#)).
-Tamiti, T. I., Joshi, B., Hasan, R., Hasan, R., Athay, T., Mamun, N., & Barua, A. (2025). *A high-fidelity speech super resolution network using a complex global attention module with spectro-temporal loss* ([arXiv:2507.00229](#)).
-Prome, R. T., Tamiti, T. I., & Barua, A. (2025). *Leveraging the potential of prompt engineering for hate speech detection in low-resource languages* ([arXiv:2506.23930](#)).
-Tamiti, T. I., & Barua, A. (2025). *A practical approach to power saving in hearables using sub-Nyquist sampling with bandwidth extension* ([arXiv:2506.22321](#)).
-Hemel, R. I., Asef, M., & Tamiti, T. I. (2024). Enhancing diagnostic accuracy in Bangla: LLM approaches to gender based medical specialty prediction. In 2024 27th International Conference on Computer and Information Technology (**ICCIT**) (pp. 873–878). **IEEE**.

RESEARCH EXPERIENCE

I am currently conducting research under [Dr. Anomadarshi Barua](#) at the [System Design and Security Lab](#), George Mason University. Our work focuses on developing Generative AI models that can reconstruct high-frequency, intelligible audio from ultra-low-frequency vibration data collected by sensors. This research has wide-ranging applications in system security, assistive medical technologies, and surveillance.

WORK EXPERIENCE

-Graduate Research Assistant: System Design and Security Lab, George Mason University
-Graduate Teaching Assistant: Department of Cyber Security and Engineering, George Mason University
-Courses: a) CYSE-580 Hardware and Cyber Physical Systems Summer 2025, Spring 2025, Fall 2024
b) CYSE-476 Cryptography Fall 2024, Fall 2025
c) CYSE-683 Reverse Engineering Industrial Automation Spring 2026
-Lecturer: Department of Computer Science & Engineering, North Bengal International University, Rajshahi, Bangladesh Spring, 2023
-Courses Taught: a) Computer Graphics, b) Peripherals and Interfacing, and c) Introduction to Information, Communication and Technology

-Internship: Monstarlab Bangladesh, An offshore center of Monstarlab Japan, a global end-to-end enterprise digital solutions provider April 2021- June 2021

- Responsibilities: Client and server-side web application development, Android mobile application development, UI/UX design & Design patterns.

Standardized Test Scores

- Classroom English Language Teaching Demonstration (CELTD) test passed May 2025
- GRE: Total 311, Quant 163, Verbal 148, AWA- 3 November 2023
- TOEFL: Total 93, Reading 24, Listening 25, Speaking 23, Writing 21 January 2023

TECHNICAL SKILLS

- Languages: Deep learning, Machine learning, Python, R, C, C++, MATLAB, CSS, HTML
- Packages/Libraries: Librosa, Torch, Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, Keras, TensorFlow, GPT's API, Hugging Face's libraries
- Database Management: MySQL XAMPP

AWARDS

- The Prime Minister's Gold Medal awarded by the Prime Minister of Bangladesh for securing the first place in the Faculty of ECE, Rajshahi University of Engineering & Technology (RUET) (to be awarded)
- The University Gold Medal awarded by the President of Bangladesh for securing the first place in the Department of ECE, Rajshahi University of Engineering & Technology (RUET) (to be awarded)
- Student of the Year for the highest GPA among 57 students in the department during the Freshmen, Sophomore, Junior, and Senior years, Rajshahi University of Engineering & Technology (RUET)
- Excellent Leadership Award for representing the class during the Junior year, Rajshahi University of Engineering & Technology (RUET)
- Duke of Edinburgh's International Award Recipient of The Bronze Standard, May 2016
- UGC Merit Scholarship Recipient of \$250 for achieving highest GPA, senior year at RUET, May 2023

RELEVANT COURSES COMPLETED

Digital Signal Processing, Hardware & Cyber Physical Systems, Cyber Security System Engineering, Cyber Security Engineering Fundamentals, Seminar, Operating Systems, Neural Networks & Fuzzy Systems, Digital Image Processing, Software Engineering & Information System Design, Data Structure and Algorithms, Database Systems, Object Oriented Programming, Computer Networks, VLSI Design, Electromagnetic Fields & Waves, Analog Electronics I & II, Industrial Electronics, Computer Aided Instrumentations, and Communication Engineering

REFERENCES

Dr. Anomadarshi Barua

Assistant Professor, Department of Cyber Security and Engineering

George Mason University, Virginia, USA

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Sagor Chandro Bakchy

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Rajshahi University of Engineering & Technology, Bangladesh

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